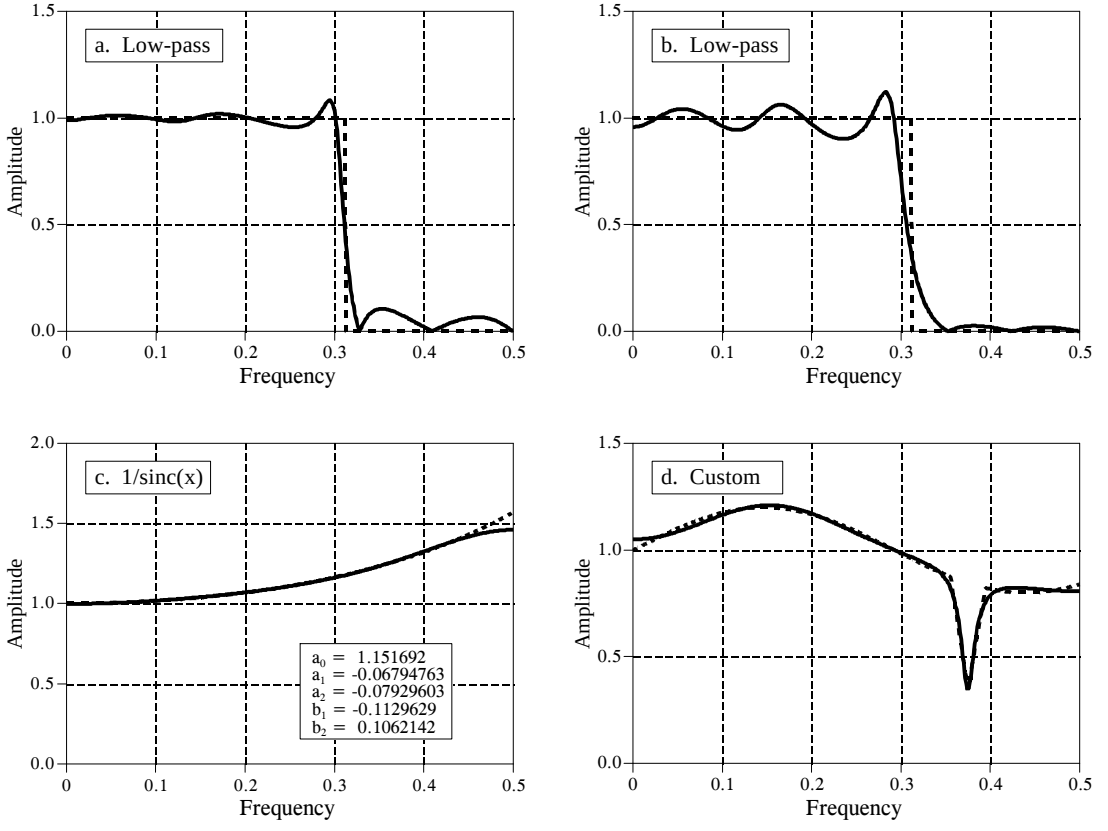

FIGURE 26-13
Iterative design of recursive filters. Figure (a) shows an 8 pole low-pass filter with the error equally distributed between 0 and 0.5. In (b), the error has been weighted to force better performance in the stopband, at the expense of error in the passband. Figure (c) shows a 2 pole filter used for the 1/sinc(x) correction in digital-to-analog conversion. The frequency response in (d) is completely custom. In each figure, the desired frequency response is shown by the dotted line, and the actual frequency response by the solid curve.

frequency response of the designed filter. Each of these filters requires several minutes to converge on a 100 MHz Pentium. Figure (a) is an 8 pole low-pass filter, where the error is equally weighted over the entire frequency spectrum (the program as written). Figure (b) is the same filter, except the error in the stopband is multiplied by *eight* when ER is being calculated. This forces the filter to have less stopband ripple, at the expense of greater ripple in the passband.

Figure (c) shows a 2 pole filter for: $1/sinc(x)$. As discussed in Chapter 3, this can be used to counteract the zeroth-order hold during digital-to-analog conversion (see Fig. 3-6). The error in this filter was only summed between 0 and 0.45, resulting in a better match over this range, at the expense of a worse match between 0.45 and 0.5. Lastly, (d) is a very irregular 6 pole frequency response that includes a sharp dip. To achieve convergence, the recursion coefficients were initially set to those of a notch filter.